

MainCrafts Internship – Task 1 Report

House Price Prediction Using Linear Regression

Intern Details

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****Domain:** AI/ML Internship**

****Task:** House Price Prediction Using Linear Regression**

Project Objective

The objective of this project was to understand and implement Linear Regression on a real-world housing dataset. The project involved loading and exploring the dataset, performing data analysis, building a machine learning model, evaluating its performance, and documenting the complete workflow.

Day 1 Progress

On the first day, I focused on understanding the basics of Linear Regression and setting up the project environment.

I created a GitHub repository to maintain the project and track my progress throughout the internship. After organizing the project files, I extracted and examined the provided California Housing dataset.

During the setup process, I faced challenges related to Python environments, package installation, dataset paths, and file loading. After troubleshooting these issues, I successfully configured the environment and loaded the dataset into a Pandas DataFrame.

The following initial analyses were performed:

- * Displayed sample records using `df.head()`
- * Checked dataset dimensions using `df.shape`
- * Examined data types using `df.info()`
- * Generated statistical summaries using `df.describe()`
- * Checked for missing values using `df.isnull().sum()`

By the end of Day 1, the project environment was fully set up and the dataset was successfully loaded and explored.

Day 2 Progress

On the second day, I focused on understanding the dataset in greater detail.

I analyzed the available features and their statistical properties to understand the factors that may influence housing prices. The dataset contained information related to median income, house age, room counts, population, households, latitude, and longitude.

I verified that the dataset did not contain missing values and began identifying patterns that could be useful during model development.

Key observations:

- * Median income appeared to have a strong relationship with house prices.
- * Geographical features such as latitude and longitude may influence housing values.
- * The dataset contained a sufficient number of observations for model training.

At the end of Day 2, I had a clear understanding of the dataset structure and was ready to move towards model development.

Day 3 Progress

On the third day, I implemented the Linear Regression model.

The dataset was divided into training and testing sets using an 80:20 split ratio. The Linear Regression algorithm was trained using the training dataset, and predictions were generated for the test dataset.

The model was evaluated using standard machine learning performance metrics.

Evaluation Results

- * Mean Absolute Error (MAE): 50,413.99
- * Mean Squared Error (MSE): 4,801,699,798.13
- * Root Mean Squared Error (RMSE): 69,294.30
- * R^2 Score: 0.642

The R^2 score indicates that the model was able to explain approximately 64.2% of the variation in house prices using the available features.

This stage helped me understand the complete machine learning workflow, including data splitting, model training, prediction generation, and performance evaluation.

Day 4 Progress

On the fourth day, I focused on project completion, visualization, documentation, and final review.

I generated visualizations to better understand the dataset and model behavior, including:

- * Feature distribution plots
- * Correlation analysis
- * Actual vs Predicted value comparison

I also reviewed the notebook structure, removed unnecessary debugging code, organized the workflow, and documented important observations and conclusions.

The project repository was updated on GitHub with all completed work and documentation.

Key Learnings

Throughout this project, I learned:

- * Fundamentals of Linear Regression
- * Data loading and preprocessing using Pandas
- * Exploratory Data Analysis (EDA)
- * Data visualization using Matplotlib and Seaborn
- * Train-test splitting techniques
- * Model evaluation using MAE, MSE, RMSE, and R^2 Score
- * Git and GitHub workflow for project management
- * Troubleshooting environment and dependency issues

Challenges Faced

During the project, I encountered several practical challenges:

- * Python package installation issues
- * Multiple Python environment conflicts
- * Dataset file path errors
- * Encoding-related dataset loading issues

Resolving these issues improved my debugging and problem-solving skills and provided valuable real-world experience.

Conclusion

This project successfully demonstrated the implementation of a Linear Regression model for house price prediction using the California Housing Dataset.

The model achieved an R^2 score of 0.642, indicating a reasonable predictive performance for a baseline Linear Regression approach. Through this project, I gained practical experience in the complete machine learning workflow, including data exploration, preprocessing, model training, evaluation, documentation, and version control using GitHub.

This internship task helped strengthen my understanding of machine learning concepts and provided hands-on exposure to real-world project development practices.

Project Repository

GitHub Repository:

<https://github.com/khasif18/house-price-predictor->

Thank you to MainCrafts for providing this learning opportunity and practical exposure to machine learning project development.